

Data Availability Statement

For the manuscript: “Systematic screening of brain neurotransmitter enzymes for radical pair mechanism competence reveals quantitative constraints on magnetic field sensitivity”

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Data availability statement

Software and code

The complete computational pipeline used to generate all results in this work is released as the open-source Python package `qbsscreen` (v0.1.0), distributed under the MIT License. The package provides a unified command-line interface for reproducing every numerical result, figure, and table presented here.

- **Source code repository:** https://github.com/deeptell-inc/brain_protein_screening
- **Package distribution (PyPI):** `pip install qbsscreen`
- **License:** MIT (permissive open source)
- **Persistent identifier:** A versioned snapshot (v0.1.0) will be deposited on Zenodo upon acceptance and assigned a DOI of the form `10.5281/zenodo.XXXXXXX` for long-term archival and citation.
- **Programming language:** Python (≥ 3.9)
- **External dependencies (version-pinned in `pyproject.toml`):**
 - GFN2-xTB v6.7.1 (Grimme group, free for academic use)
 - Open Babel v3.1.0 (GPL-2.0)
 - PySCF v2.12.1 (Apache 2.0; required for DFT/CASSCF modules)
 - NumPy ≥ 1.24 , SciPy ≥ 1.10 , Matplotlib ≥ 3.7 , Biopython ≥ 1.80

Reproducibility

Each result claim in this paper is automatically verified by an included test suite (`pytest qbsscreen/tests/`). The 20 tests cover:

- **Spin Hamiltonian construction** (Pauli operator algebra, singlet projector trace, dimension scaling)
- **Magnetic field effect** (MAO-A MFE -4.4% at $J = 0$; vanishing MFE at $|J| > 500$ MHz)
- **Spin relaxation** (r^{-6} PRE scaling, strong-coupling condition $A/\omega_S > 100$, ZQ TDM $= 0.7 \mu_B$)
- **CRY simulation** (asymmetric HFC distribution, larger MFE than MAO-A)
- **Paper numbers** (3.2 ms diamagnetic T_2 , 64-fold PLP/FAD nuclear T_2 gap, CASSCF $y_0 < 0.1$, 2.7–8.2 min circadian shift)
- **Full screening pipeline** on cryptochrome (PDB 4I6G): site detection, FAD TDM (~ 10 D), SOC ($\sim 67 \text{ cm}^{-1}$), $n_P = 2$, T_2^* , T_1^e

The tests run in ~ 7 seconds on a standard workstation and verify all 11 quantitative claims highlighted in the main text within their physical tolerance ranges.

Reproduction of paper results from the command line

```
# 1. Screen all 9 brain proteins (Table 1, Fig. 1)
for pdb in 4I6G 2BXR 1OJA 2DU8 3RBF 2OKK \
          3L6B 1TOL 1I10; do
    qbscreen screen $pdb -o results_${pdb} \
                  --no-hess
done

# 2. RPM spin dynamics (Fig. 4, ED Fig. 6, 8)
qbscreen simulate

# 3. DFT/TDDFT validation (ED Table 3)
qbscreen dft-validate

# 4. CASSCF(2,2)/NEVPT2 (ED Fig. 7)
qbscreen casscf

# 5. Reaction coordinate scan (ED Fig. 7a)
qbscreen reaction-scan

# 6. Verify reproducibility
pytest qbscreen/tests/ -v
```

Datasets

Input data (publicly available). All input crystal structures were obtained from the RCSB Protein Data Bank (<https://www.rcsb.org>) under the following accession codes:

- PDB [4I6G](#) — *Drosophila melanogaster* cryptochrome (CRY)
- PDB [2BXR](#) — human monoamine oxidase A (MAO-A)
- PDB [1OJA](#) — human monoamine oxidase B (MAO-B)
- PDB [2DU8](#) — human D-amino acid oxidase (DAO)
- PDB [3RBF](#) — human aromatic L-amino acid decarboxylase (DDC)
- PDB [2OKK](#) — human glutamate decarboxylase (GAD67)
- PDB [3L6B](#) — human serine racemase (SRR)
- PDB [1TOL](#) — human isocitrate dehydrogenase (IDH)
- PDB [1I10](#) — human lactate dehydrogenase (LDH)

Computed data (archived in repository). The following derived datasets are included in the `qbscreen` repository under `data/` and `results/`:

- Optimised quantum-active site geometries (87–156 atoms each, XYZ format)
- Cofactor-specific HFC tensors and SOC constants (JSON)
- Five-channel T_2^e decompositions for all 9 proteins (JSON)
- Paramagnetic relaxation enhancement (PRE) curves vs r (CSV)
- RPM singlet yield $\Phi_S(B)$ on a 40×16 field-exchange grid (HDF5)
- CASSCF(2,2)/NEVPT2 wavefunctions and CI coefficients at three reaction-coordinate geometries (PySCF `chkfile` format)

- Vibrational analysis output for top-ranked sites (xtb format)
- All raw data underlying every figure and table (CSV/JSON)

Format and access. All data are released in non-proprietary formats (XYZ, JSON, CSV, HDF5) and are immediately accessible without restriction. No personally identifiable, ethically restricted, or commercially sensitive information is involved.

Underlying experimental data

This work is a purely computational study; no new experimental data were generated. Experimental values used for benchmarking and validation are cited in the main text from the original sources (Weber *et al.*[ref] for ^{31}P ENDOR; Islam *et al.*[ref] for FAD oscillator strengths; Maeda *et al.*[ref] for cryptochrome MFEs; Xu *et al.*[ref] for CRY magnetic sensitivity).

Hardware and computational environment

All calculations were performed on a single workstation: Apple M4 Max (16 cores, ARM64), 64 GB RAM, macOS 26.2. No GPU acceleration was used. Total wall time for full reproduction: ~ 30 minutes (xTB screening) + ~ 20 minutes (spin dynamics) + ~ 30 minutes (DFT/TDDFT) + ~ 8 hours (CASSCF/NEVPT2 at 3 geometries). Container images (Docker/Singularity) with all dependencies pre-installed will be released alongside the v1.0 package version.

Conditions of access and reuse

The MIT License permits unrestricted use, modification, and redistribution of the software, including for commercial purposes, with attribution. Citation of this paper is requested for academic use. Issues, feature requests, and contributions are welcome via the GitHub repository.

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